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Ben Smyth and Alfredo Pironti, *INRIA Paris-Rocquencourt*

We identify logical web application flaws which can be exploited by TLS truncation attacks to desynchronize the user- and server-perspective of an application's state. It follows immediately that servers may make false assumptions about users, hence, the flaw constitutes a security vulnerability. Moreover, in the context of authentication systems, we exploit the vulnerability to launch the following practical attacks: we exploit the Helios electronic voting system to cast votes on behalf of honest voters, take full control of Microsoft Live accounts, and gain temporary access to Google accounts.

Ben Smyth, INRIA Paris-Rocquencourt

Alfredo Pironti, INRIA Paris-Rocquencourt

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